The skills of induction

- Recognizing that induction might apply
- Formulating an appropriate predicate
- Deriving base case and inductive step
- Modifying our predicate if neccesary

Recall

From §2.4, we were told:

($\forall n \geq 1$, is implied when there is a free variable such as n) $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$

- Induction will not help us learn where the formula comes from
- It will help us verify if the formula is correct

Syntacting matching suggets:

let $P(n)$ be "$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$"

No guarantee yet, that this predicate will "play nice" with induction

Base Step:

$P(1)$ is the statement

"$\sum_{k=1}^{1} k^2 = \frac{1(1+1)(2(1)+1)}{6}$"

$1^2 = \frac{1 \times 2 \times 3}{6}$

or $1 = 1$

$P(1)$ is true.

Focus on the inductive step

We want to show

$P(k) \to P(k+1)$

For generic k

We can treat $P(k)$ as a hypothesis

and then try to prove

$P(k+1)$

$P(k)$ is the claim:

$\sum_{i=1}^{k} i^2 = \frac{k(k+1)(2k+1)}{6}$

Strategically all the k's have been renamed to i's

We want to show $P(k+1)$ i.e

$\sum_{i=1}^{k+1} i^2 = \frac{(k+1)(k+2)(k+3)}{6}$

Left Hand Side = $\sum_{i=1}^{k+1} i^2 = \left(\sum_{i=1}^{k} i^2\right) + (k+1)^2$

$= \frac{k(k+1)(2k+1)}{6}$ + (k + 1)^2

Replacing equal quantities appealing to our assumption.

Algebra Shows:

$\frac{(k+1)(k+2)(k+3)}{6} = \frac{k(k+1)(2k+1)}{6} + (k+1)^2$

$= \frac{(k+1)(k+2)(2k+3)}{6} + \text{Right Hand Side}$

Since LHS of $P(k+1) =$ RHS of $P(k+1)$

$P(k+1)$ must be true, provided $P(k)$ was.

We showed $P(k) \to P(k+1)$ thus

$\forall k \geq 1 (P(k) \to P(k+1))$

Warning

THe base case matters. If we tried

$Q(n)$ is $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6} + 4$

We could still prove $Q(k) \to Q(k+1)$

But $Q(1)$ wouldve been

$1^2 = 1 + 4$ which is false.

Recall the Fibonacci sequence

$f_0 = 0, f_1 = 1$ and $f_{n+2} = f_{n+1} + f_n$ if $n \geq 0$

It starts:

$0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144$

f is just a function, so we can ask about its $\mathcal{O}$ behavior

Claim

f_n is $\mathcal{O}(2^n)$

First Guess

Can we use the predicate

$P(n)$ given by $f_n \leq 2^n$?

Rough Work

We want to conclude $P(k+1)$

$f_{k+1} \leq 2^{k+1}$

Now

(This would be true if $f_k \leq 2^k$ and $f_{k-1} \leq 2^{k-1}$ i.e $P(k) \wedge P(k-1)$)

$f_{k+1} = f_k + f_{k-1} \leq 2^k + 2^{k+1}$

$= \frac{1}{2} 2^{k+1} + \frac{1}{4} 2^{k+1}$

$= \frac{3}{4} 2^{k+1} \leq 2^{k+1}$

We sucessfuly showed $P(k) \wedge P(k-1) \to P(k+1)$

(Wrong format for induction)

Clever Fix

Use an easier induction to prove $f_{k-1} \leq f_k \mid \forall k$ then

$f_{k+1} = f_k + f_{k-1} \leq f_k + f_k = 2f_k \leq 2 * 2^k = 2^k + 1$

Now we have $P(k) \to P(k+1)$

Systematic Fix

Let $Q(k)$ be $P(k) \wedge P(k-1)$ we've shown

$Q(k) \to P(k+1)$ but also

$Q(k) \to P(k)$ (Simplifcation)

so

$Q(k) \to P(k+1) \wedge P(k)$

Equivalently $Q(k) \to Q(k+1)$

In terms of Q our vase case is

$Q(1)$ which is $P(1) \wedge P(0)$

which is

$f_1 \leq 2 \wedge f_0 \leq 1$

which is true, since $1 \leq 2 \wedge 0 \leq 1$

We conclude

$\forall n \geq 1 Q(n)$ thus

$\forall n \geq 1 P(n) \wedge P(n-1)$

With consequence

$\forall n \geq 1 P(n)$

Literally

$\forall n \geq 1, f_n \leq 2^n$ so

f_n is $\mathcal{O}(2^n)$

with $k = 0$ and $C = 1$ as witnesses

Observation

f_n is $\Omega\left(\left(\frac{3}{2}\right)^n\right)$

The right predicate, for our conclusion is

$P(n)$ given by $f_n \geq \left(\frac{3}{2}\right)^n$

We can show

$f_{n+1} = f_n + f_{n-1}$

$\geq \left(\frac{3}{2}\right)^n + \left(\frac{3}{2}\right)^{n-1}$

$= \frac{2}{3}\left(\frac{3}{2}\right)^{n+1} + \frac{9}{4}\left(\frac{3}{2}\right)^{n-1}$
$= \frac{10}{9}\left(\frac{3}{2}\right)^{n+1} \geq \left(\frac{3}{2}\right)^{n+1}$
Thus we showed $P(n) \wedge P(n-1) \rightarrow P(n+1)$

The related predicate $Q(n)$ given by $Q(n)$ is $P(n) \wedge P(n-1)$

has the property

$Q(k) \rightarrow Q(k+1)$

BUT

$f_0 = 0 < \left(\frac{3}{2}\right)^0$ so $P(0)$ is false

and $Q(1)$ is false

$n, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12$

$f_n, 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144$

$\left(\frac{3}{2}\right)^n$
We find $\left(\frac{3}{2}\right)^{10} > 55$
$\left(\frac{3}{2}\right)^{11} < 89$
$\left(\frac{3}{2}\right)^{12} < 144$
$Q(4)$ is true, but nothing smaller

Since

$Q(11)$

$\forall k \geq 1 (Q(k) \rightarrow Q(k+1))$

We conclude

$\forall n \geq 11 Q(n)$

thus

$\forall n \geq 11 P(n)$

so $f_n \geq \left(\frac{3}{2}\right)^n$ is $n \geq 11$

we get f_n is $\Omega\left(\left(\frac{3}{2}\right)^n\right)$ with witnesses $k = 11$ and $C = 1$

If we want to show f_n is $\mathcal{O}(C^n)$

we can.

The inductive step works any times

$C > \frac{1+\sqrt{5}}{2}$

The base case will take trial and error.